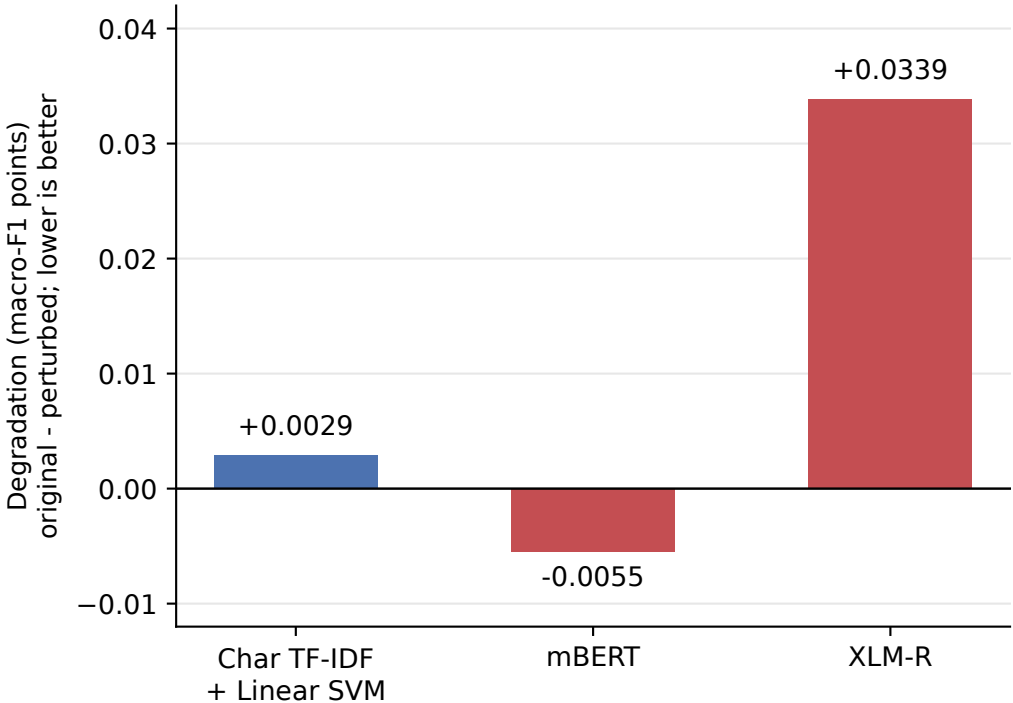

Mean over 10 synthetic perturbation types, paired test messages. Negative = perturbed macro-F1 slightly higher than original (not interpreted as a robustness gain). Not a measure of real-world adversarial robustness.